

BrewTroller PID Display

Kit Assembly Instructions

v1

1. **It is important to solder the components in the order specified in these instructions. Soldering the LED Display Modules out of order can make it IMPOSSIBLE to complete the kit!**
2. You can find the latest version of this document at <http://www.vonnieda.org/brewtroller-pid-display> along with pictures and videos to help you along.
3. Print these instructions if you have a printer and check off each one as you go along. It helps, trust me.
4. Check that your kit has all the necessary parts:
 1. (1) Printed Circuit Board
 2. (1) 10k Resistor (Brown, Black, Orange)
 3. (1) .1 uF Capacitor
 4. (8) PNP Transistors
 5. (1) ATTiny88 Micro-controller IC
 6. (1) 100 Ohm Resistor Network IC (1-101)
 7. (1) 1k Ohm Resistor Network IC (1-102)
 8. (3) 4 Pin Header
 9. (4) Dual Digit 7-Segment LED Display Modules
5. Separate your LED Display Modules by color / line.
 - Modules marked **A512RI** are **GREEN**
 - Modules marked **A514RI** are **RED**
6. We still start working from the back of the PCB, where the white lettering is.
7. Insert the ATTiny88 IC into the PCB at the position marked IC1. Make sure you notice the indentation in the marking and line it up with the indentation on the IC. You may find that the pins need to be bent a bit in to get it to fit. To do this, place the IC on it's side and gently apply pressure down and forward to bend the pins in slightly. It doesn't take much! Then flip it over and do the other side. It should slide right into the PCB holes.
Don't solder yet.
8. Insert the 100 Ohm Resistor Network IC marked 1-101 into the PCB at the position marked RN1. You may have to bend the pins like you did in step 5.
Don't solder yet.
9. Insert the 1k Ohm Resistor Network IC marked 1-102 into the PCB at the position marked RN2. You may have to bend the pins like you did in step 5 and 6.
Don't solder yet.
10. Carefully flip the board over making sure the ICs do not fall out. Put the board down on your

table and apply light pressure to the back side, pressing all the ICs in and seating them fully.

11. Solder all 28 pins of the ATTiny88, the 16 pins of the 100 Ohm Resistor Network and the 16 pins of the 1k Ohm Resistor Network. Go slow and check your work regularly. Make sure you don't miss any pins and make sure you get good solder flow on each pin.
12. Flip the board back over so you are looking at the side with the lettering again.
13. Bend one lead of the 10k Resistor (Brown, Black, Orange) down along side of the resistor so that the two leads of the resistor both point down. Look at the **[PICTURE]** in case this is confusing.
14. Insert the 10k Resistor (Brown, Black, Orange) into the PCB at the position marked R1.
15. Insert the .1 uF Capacitor into the PCB at the position marked C1.
16. Flip the board to the front (no white lettering) and slightly bend the leads of the capacitor and resistor so that they do not fall out.
17. Solder the leads of the capacitor and resistor.
18. Flip the board to the back again.
19. Detach your 8 PNP Transistors from the tape they are attached to. This is easy if you hold the tape firmly and gently rock the transistor back and forth while pulling gently. They should come out easily.
20. From the back of the board (with the white lettering) insert all 8 transistors into the positions marked T1 – T8. Make sure you notice the curved part and the flat part on the white lettering and line up the transistors as shown.
21. Flip the board again to the front and slightly bend the 2 outer leads of each transistor. This will keep them from falling out while you solder.
22. Solder all 8 transistors.
23. Flip the board to the back and insert the (2) 4 pin headers at the locations marked with SDA, SCL, +5V and GND.
24. Solder the 4 pin headers from the front of the board.
25. Use a pair of right angle cutters (wire cutters, snips, etc.) to trim all the leads of everything but the ICs. I like to trim them right down to where the solder wicks up the lead. About 2mm from the board, or 1/16".
26. Now, use a small flat head screwdriver to bend the soldered leads of the ICs away from center. It should not take much force at all. This is so that the LED Modules will sit flush against the board. Please see the **[VIDEO]** to see how this is done.

27. Now, from the front of the board (with no lettering) insert the (4) LED Display Modules. Take your time and make sure of the following things:
 - Make sure the modules are right side up. The decimal points should be towards the bottom.
 - Make sure you have put the right colors in the right lines. The only way to identify the colors is by the numbers on the side so look at the list at the top of this document and check everything twice! It is nearly impossible to fix this once you have soldered the Modules in.
 - Check to make sure all the Modules sit flush. If they do not you may have forgotten to bend the pins of the ICs in step 25.
28. Flip the board to the back and apply slight pressure to the back of the board, seating the LED Modules on the front. Check again from the sides that they are all flush.
29. Solder the (4) LED Display Modules.
30. If your BrewTroller does not have the 4 pin I2C header populated with pins you can use the 3rd included 4 pin header to do that. Just carefully insert it into your BrewTroller board and solder it from the back. The I2C header is between the USB connector and the LCD header pins.
31. Celebrate! The kit is now assembled and ready to test!
32. Power off your BrewTroller if it is powered on.
33. Attach a 4 pin female to female cable from the BrewTroller I2C header to either of the BrewTroller PID Display I2C headers. Be sure you match pin to pin. Each header is labeled SDA, SCL, +5V and GND. A straight through cable should have the same colors on each pin.
34. Power on your BrewTroller. Your BrewTroller PID Display should show BTPD on the top line and v1 on the bottom line for 2 seconds, then ADDR on the top line and 20 on the bottom line for 2 seconds and then go blank. If you do not see this turn off the power to your BrewTroller and recheck all of your solder joints and connections. If you still cannot get this to show please do not power up the board again and email me at jason@vonnieda.org.
35. Celebrate! Your kit is now tested and ready to use! Please see the Owners Manual for information about setup and usage.